

Ex. 4.2 | Ch. 4 | P115 | Chloé | 2025.7.28

解 (a) 设波函数 $\psi(x, y, z) = X(x)Y(y)Z(z)$, 代入薛定谔方程:

$$-\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi = E\psi, \text{ 又 } V=0, \text{ 得 } -\frac{\hbar^2}{2m} \left(\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} \right) = E\psi$$

$$\Rightarrow -\frac{\hbar^2}{2m} \left(\frac{X''}{X} + \frac{Y''}{Y} + \frac{Z''}{Z} \right) = E$$

$$\text{令 } -\frac{\hbar^2}{2m} \frac{X''}{X} = E_x, \quad -\frac{\hbar^2}{2m} \frac{Y''}{Y} = E_y, \quad -\frac{\hbar^2}{2m} \frac{Z''}{Z} = E_z$$

• 以 $X(x)$ 为例求解 $X'' + \frac{2mE_x}{\hbar^2} X = 0$, 又 $X(0) = X(a) = 0$, 可得:

$$X(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n_x \pi x}{a}\right), \quad E_x = \frac{\hbar^2 \pi^2 n_x^2}{2ma^2}, \quad n_x = 1, 2, 3, \dots (\text{量子数取正整数保证波函数非0}).$$

$$\text{同理: } Y(y) = \sqrt{\frac{2}{a}} \sin\left(\frac{n_y \pi y}{a}\right), \quad E_y = \frac{\hbar^2 \pi^2 n_y^2}{2ma^2},$$

$$Z(z) = \sqrt{\frac{2}{a}} \sin\left(\frac{n_z \pi z}{a}\right), \quad E_z = \frac{\hbar^2 \pi^2 n_z^2}{2ma^2}, \quad \text{因此, 三维定态波函数为:}$$

$$\psi_n(x, y, z) = \left(\frac{2}{a}\right)^{3/2} \sin\left(\frac{n_x \pi x}{a}\right) \sin\left(\frac{n_y \pi y}{a}\right) \sin\left(\frac{n_z \pi z}{a}\right)$$

$$\text{能量为 } E_{n_x n_y n_z} = \frac{\hbar^2 \pi^2}{2ma^2} (n_x^2 + n_y^2 + n_z^2) = \frac{\hbar^2}{8ma^2} (n_x^2 + n_y^2 + n_z^2)$$

$$(b) \text{ 能量 } N = n_x^2 + n_y^2 + n_z^2, \quad \textcircled{1} N=3 \text{ 时, } (n_x, n_y, n_z = 1, 1, 1) \quad E_1 = \frac{\hbar^2}{8ma^2} \cdot 3 \quad E_2 = \frac{\hbar^2}{8ma^2} \cdot 6$$

$$\text{简并度 } 1; \quad \textcircled{2} N=6 \text{ (} n_x, n_y, n_z \text{ 为 } 1, 1, 2 \text{ 排列), 共有 } 3 \text{ 种排列, 简并度为 } 3$$

$$\textcircled{3} N=9 \text{ 时, } n_x, n_y, n_z \text{ 为 } 1, 2, 2 \text{ 或 } 3, 0, 0 \text{ (舍)}, \quad E_3 = \frac{\hbar^2}{8ma^2} \cdot 9, \quad \text{简并度为 } 3,$$

$$\textcircled{4} N=11 \text{ 时, } n_x, n_y, n_z \text{ 为 } 1, 1, 3, \text{ 排列数为 } 3, \quad \text{简并度为 } 3, \quad E_4 = \frac{\hbar^2}{8ma^2} \cdot 11$$

$$\textcircled{5} N=12 \text{ 时, } n_x, n_y, n_z \text{ 为 } 2, 2, 2, \quad \text{简并度为 } 1, \quad E_5 = \frac{\hbar^2}{8ma^2} \cdot 12$$

$$\textcircled{6} N=14 \text{ 时, } n_x, n_y, n_z \text{ 为 } 1, 2, 3, \quad \text{简并度为 } 6, \quad E_6 = \frac{\hbar^2}{8ma^2} \cdot 14$$

$$(c) E_{14}, \quad N = n_x^2 + n_y^2 + n_z^2 = 14, \text{ 唯一正整数解为 } 1, 2, 3, \text{ 共有 } 6 \text{ 种排列, 简并度为 } 6$$

趣味点体现在: 简并度为 6 源于三维空间的对称性, 量子数 n 不同排列对应不同的“运动模式”, 但能量相同。相比低 N 情况, $N=14$ 的简并度跃升至 6, 体现了三维势阱中量子态随量子数增大, 因对称性带来“复杂简并结构”, 是量子力学中对称性导致简并的典型案例。